Example-3

Radial Basis Function (RBF) Kernel

$e^{-\gamma||a-b||^2}$

Input Vectors

$a = [a_1, a_2]^T \quad b = [b_1, b_2]^T$

Normal Model

Calculate $a^T b$

$a = [a_1, a_2]^T$

$b = [b_1, b_2]^T$

Mapped Vectors

Map Input data from 2D to infinite Dimens

Input: a,b

Vectors Mapping $e^{-\gamma||a-b||^2}$ a,b → u,v

Kernelized Model

Calculate $u^T v$

$U = [u_1\ u_2\ u_3\ \dots\ u_\infty$
$v_1\ v_2\ v_3\ \dots\ v_\infty]$

Higher Computational Complexity !!!

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